Chloe Shuqing Xu

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EDUCATION

Monash University **Jul 2025 - Jun 2028**
Bachelor of Computer Science | Faculty of Information Technology

EXPERIENCE

Database Development & Data Analysis Volunteer **Jun 2025 - Nov 2025**
BigRig Mover | Community Technical Service Project

- Contributed to a volunteer technical service project for transport operations, designing an **Oracle SQL** database for quotations, jobs and personnel with relational links and integrity constraints.
- Built business reports using **multi-table joins, subqueries and aggregation** to analyse quotations, staff assignments and vehicle use, handling NULL values and unmatched records.
- Supported operational improvements associated with **23% year-over-year growth in customer bookings** and an approximately **47% reduction in business data processing time**.

PROJECTS

Superstore Economic Performance Dashboard **Mar 2026 - May 2026**
R, R Shiny, tidyverse, Plotly | Data Integration & Business Analysis

- Integrated **Kaggle Superstore sales** with three external datasets covering US state GDP, population and per-capita income to analyse regional performance from **2011 to 2014**.
- Built an interactive **R Shiny dashboard** with region, year and economic-factor filters to explore sales trends, profit margins, regional comparisons and economic relationships.
- Identified the West's **31.6% share of sales** and high-sales, negative-profit markets such as Texas to support performance assessment and anomaly investigation.

Farm & Roguelike Game **Jul 2026 - Oct 2026**
Java, OOP, REST API, SOLID, GitLab | Team Project

- Contributed trading-system and **Second Spring** features to a team-developed farming and Roguelike game, applying object-oriented design to extensible gameplay mechanics.
- Worked within a game architecture that integrates **Open-Meteo seven-day forecasts** and uses rainfall, temperature and wind data to drive environmental disaster mechanics.
- Applied interfaces, polymorphism and the **single-responsibility principle** to separate disaster effects, recovery behaviour and terrain interactions.

Campus Exploration & Reward Management **Mar 2026 - May 2026**
Python, Graph Algorithms, Binary Search Trees, Max Heaps

- Modelled campus locations and weighted connections as a graph, using **recursive traversal and greedy strategies** to explore routes, calculate connection difficulty and evaluate rewards.
- Used **BSTs, max heaps and hash tables** for range queries, Top-K reward retrieval and dynamic updates; maintained leaderboard order with **merge sort**.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C++, HTML
Libraries & Databases: pandas, NumPy, R Shiny, Oracle Database, MongoDB
Tools & Technologies: Excel, Power BI, Power Automate, GitLab, Linux, REST API, JSON